

Physical Chemistry (I)

Lecture 16

2차 중간고사 안내 및 복습



2차 중간고사 안내

- 시간: 5월 17일(수) 오후 1:30~2:30
- 장소: 화학관 315호(홀수 학번), 112호(짝수 학번)
- 신분증 지참(모바일 학생증도 가능)
- 계산기 불필요

시험 범위

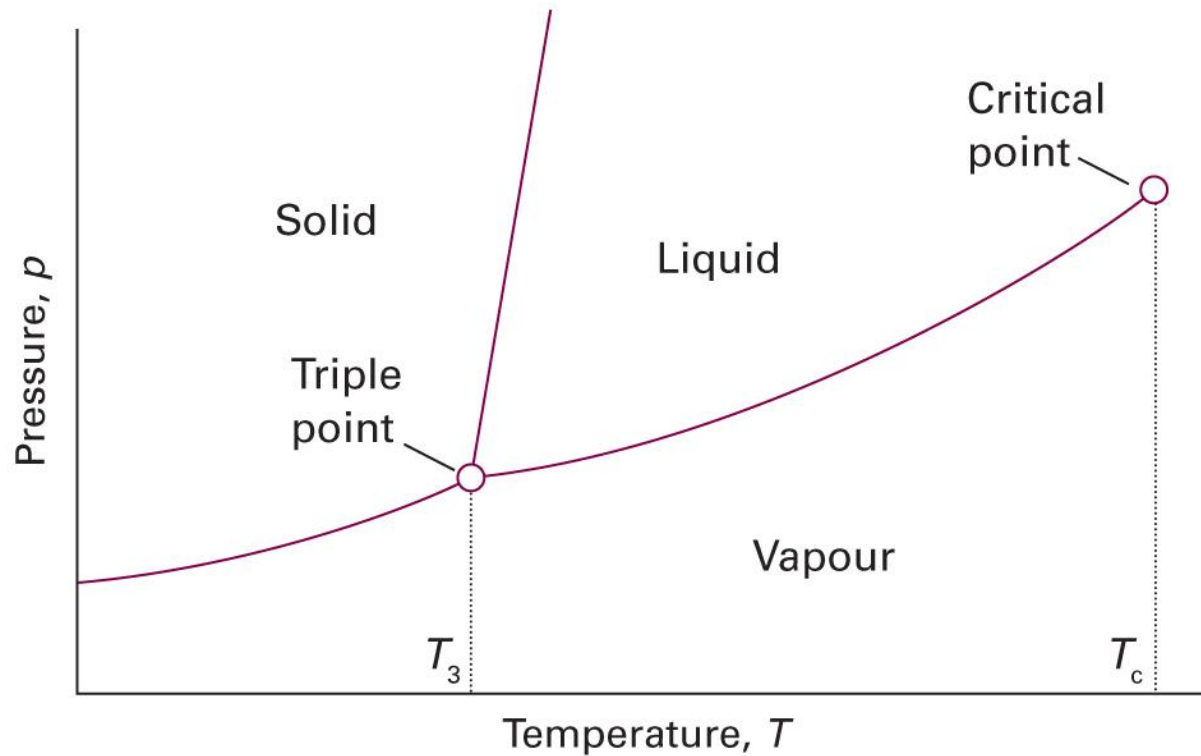
- 4장 “엔트로피” 1차 중간고사 범위 이후부터
- 5장 “헬름홀츠 에너지와 기브스 에너지”
- 6장 “통계역학”
- 7장 “상 전이”

시험 문제

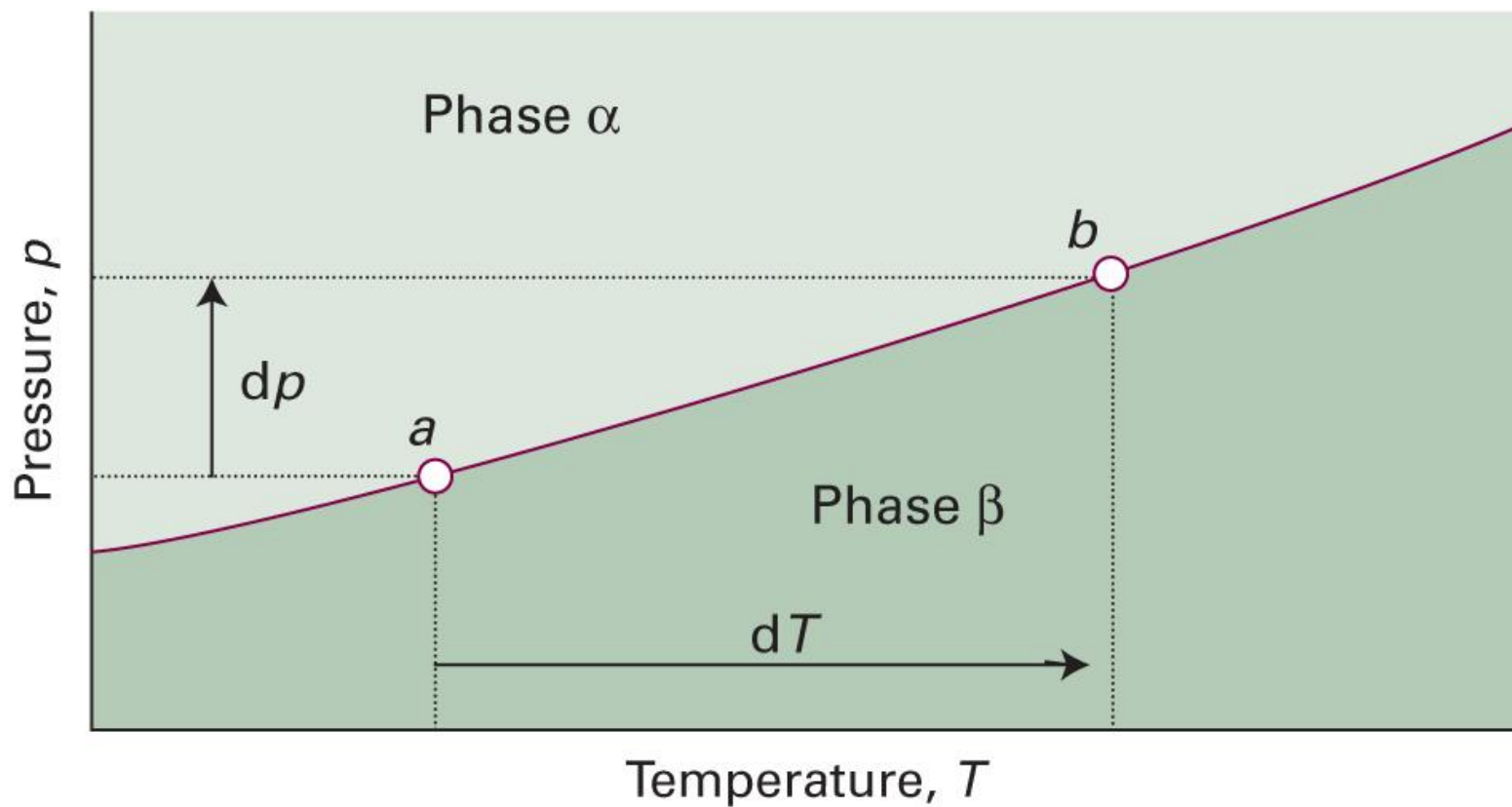
총 여섯 문제

- 과제 문제 혹은 그 변형(3문제)
 - 유도라면 중간 단계를 주지 않을 수도 있음
- 개념 설명 및 공식 유도(3문제)
 - 강의 자료 기준

Typical Phase Diagram



Phase Boundary



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4B.4)

At Phase Boundary

When two phases α and β are at equilibrium,

$$\mu(\alpha; p, T) = \mu(\beta; p, T)$$

For infinitesimal changes dp and dT ,

$$d\mu(\alpha) = d\mu(\beta)$$

Since $d\mu = V_m dp - S_m dT$ (Gibbs-Duhem equation),

$$V_m(\alpha)dp - S_m(\alpha)dT = V_m(\beta)dp - S_m(\beta)dT$$

Clapeyron Equation

$$V_m(\alpha)dp - S_m(\alpha)dT = V_m(\beta)dp - S_m(\beta)dT$$

Rearrangement leads to

$$\{S_m(\beta) - S_m(\alpha)\}dT = \{V_m(\beta) - V_m(\alpha)\}dp$$

$$\Delta_{\text{trs}}SdT = \Delta_{\text{trs}}Vdp$$

Thus,

$$\frac{dp}{dT} = \frac{\Delta_{\text{trs}}S}{\Delta_{\text{trs}}V}, \quad \frac{dT}{dp} = \frac{\Delta_{\text{trs}}V}{\Delta_{\text{trs}}S}$$

(Almost Always) Positive Slope

$$\frac{dp}{dT} = \frac{\Delta_{\text{trs}}S}{\Delta_{\text{trs}}V}$$

For a phase transition, $\Delta_{\text{trs}}S = \Delta_{\text{trs}}H/T$, and

$$\frac{dp}{dT} = \frac{\Delta_{\text{trs}}H}{T\Delta_{\text{trs}}V}$$

Hence, dp/dT is almost always positive

(unless $\Delta_{\text{trs}}H$ and $\Delta_{\text{trs}}V$ have different signs)

Solid-Liquid Boundary

$$\frac{dp}{dT} = \frac{\Delta_{\text{fus}}H}{T\Delta_{\text{fus}}V}$$

Assuming $\Delta_{\text{fus}}H$ and $\Delta_{\text{fus}}V$ are constant,

$$dp = \frac{\Delta_{\text{fus}}H}{\Delta_{\text{fus}}V} \frac{1}{T} dT$$

$$\int_{p_1}^{p_2} dp = \frac{\Delta_{\text{fus}}H}{\Delta_{\text{fus}}V} \int_{T_1}^{T_2} \frac{1}{T} dT$$

Solid-Liquid Boundary

$$\int_{p_1}^{p_2} dp = \frac{\Delta_{\text{fus}} H}{\Delta_{\text{fus}} V} \int_{T_1}^{T_2} \frac{1}{T} dT$$

After integration,

$$p_2 - p_1 = \frac{\Delta_{\text{fus}} H}{\Delta_{\text{fus}} V} \ln \left(\frac{T_2}{T_1} \right)$$

Solid-Liquid Boundary

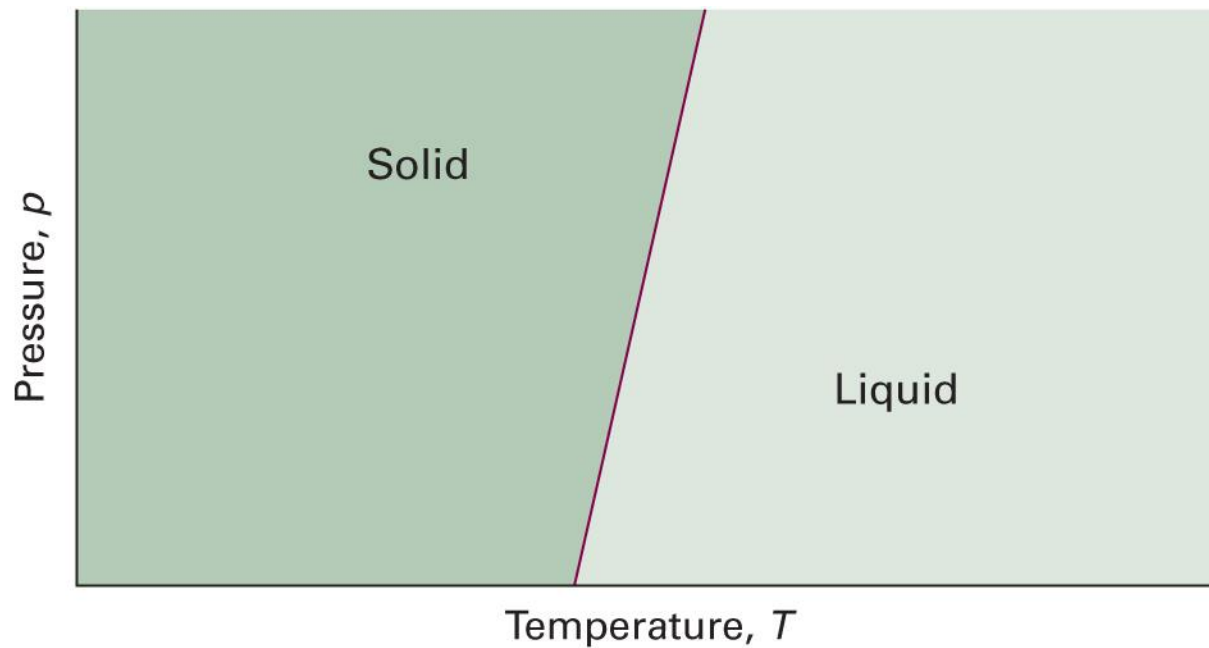
$$p_2 - p_1 = \frac{\Delta_{\text{fus}}H}{\Delta_{\text{fus}}V} \ln \left(\frac{T_2}{T_1} \right)$$

If $T_1 \approx T_2$, using $\ln(1 + x) = x - x^2/2 + \dots$,

$$\ln \left(\frac{T_2}{T_1} \right) = \ln \left(1 + \frac{T_2 - T_1}{T_1} \right) \approx \frac{T_2 - T_1}{T_1}$$

$$p_2 - p_1 = \frac{\Delta_{\text{fus}}H}{T_1 \Delta_{\text{fus}}V} (T_2 - T_1)$$

Solid-Liquid Boundary



Liquid-Gas Boundary

$$\frac{dp}{dT} = \frac{\Delta_{\text{vap}}H}{T\Delta_{\text{vap}}V}$$

Since $V_{\text{m}}(g) \gg V_{\text{m}}(l)$,

$$\Delta_{\text{vap}}V = V_{\text{m}}(g) - V_{\text{m}}(l) \approx V_{\text{m}}(g) \approx RT/p$$

Thus,

$$\frac{1}{p} \frac{dp}{dT} = \frac{\Delta_{\text{vap}}H}{RT^2}$$

(Clausius-Clapeyron equation)

Liquid-Gas Boundary

$$\frac{1}{p} dp = \frac{\Delta_{\text{vap}}H}{RT^2} dT$$

Assuming $\Delta_{\text{vap}}H$ is constant,

$$\int_{p_1}^{p_2} \frac{1}{p} dp = \frac{\Delta_{\text{vap}}H}{R} \int_{T_1}^{T_2} \frac{1}{T^2} dT$$

$$\ln \left(\frac{p_2}{p_1} \right) = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

Liquid-Gas Boundary

$$\ln \left(\frac{p_2}{p_1} \right) = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

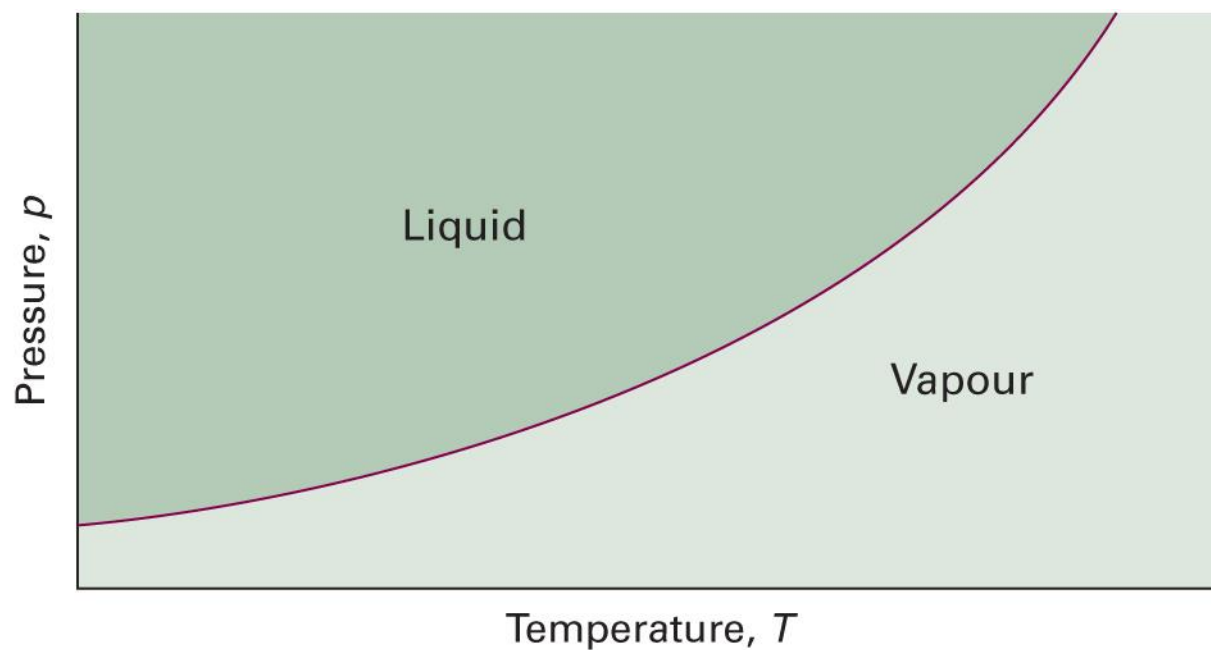
Hence,

$$p_2 = p_1 e^{A-B/T_2}$$

where

$$A = \frac{\Delta_{\text{vap}}H}{RT_1}, \quad B = \frac{\Delta_{\text{vap}}H}{R}$$

Liquid-Gas Boundary



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4B.6)

Solid-Gas Boundary

$$\frac{dp}{dT} = \frac{\Delta_{\text{sub}}H}{T\Delta_{\text{sub}}V}$$

Similarly, $\Delta_{\text{sub}}V \approx V_{\text{m}}(g) \approx RT/p$, and

$$\frac{1}{p} \frac{dp}{dT} = \frac{\Delta_{\text{sub}}H}{RT^2}$$

Assuming $\Delta_{\text{sub}}H$ is constant,

$$\ln \left(\frac{p_2}{p_1} \right) = \frac{\Delta_{\text{sub}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

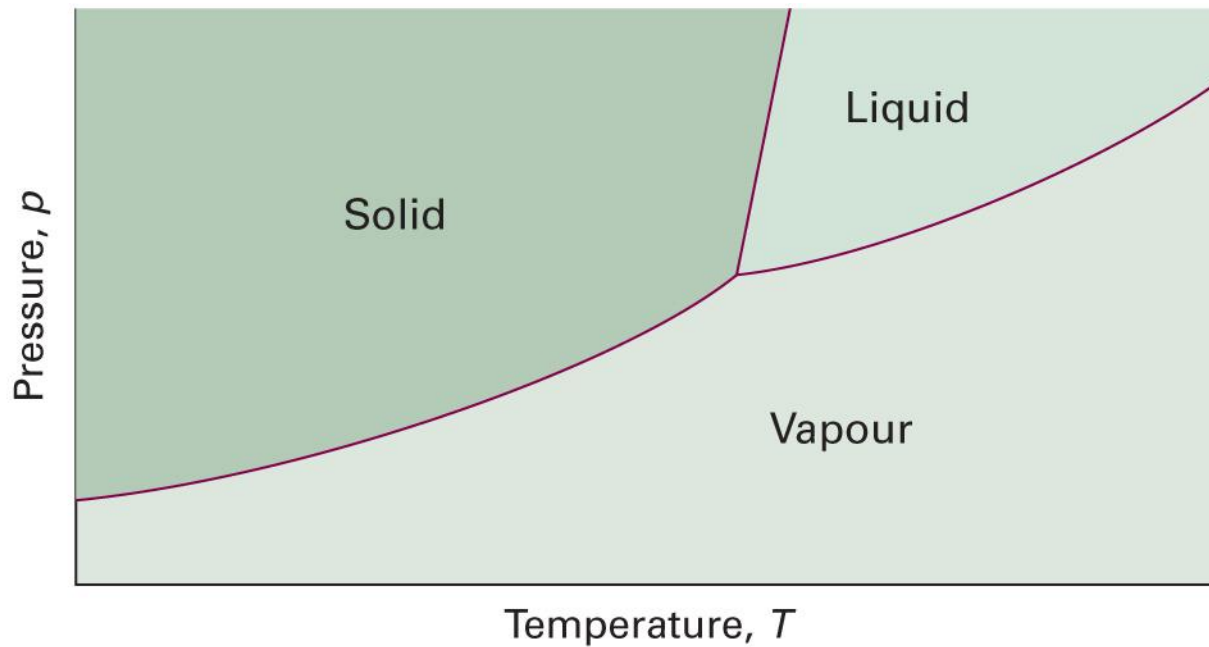
Solid-Gas vs. Liquid-Gas

$$\text{Solid-gas: } \ln\left(\frac{p_2}{p_1}\right) = \frac{\Delta_{\text{sub}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\text{Liquid-gas: } \ln\left(\frac{p_2}{p_1}\right) = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

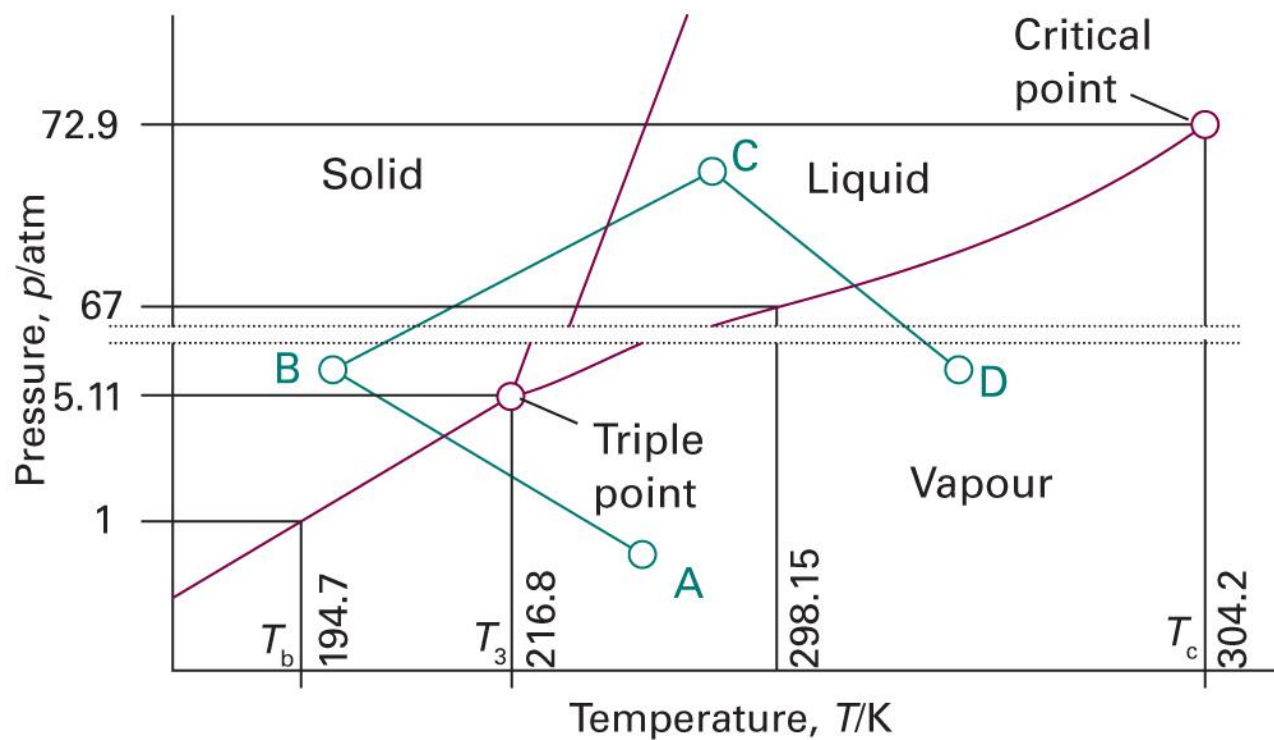
Since $\Delta_{\text{sub}}H = \Delta_{\text{fus}}H + \Delta_{\text{vap}}H > \Delta_{\text{vap}}H$, the **sublimation curve** has a **steeper slope** than the **vaporization curve**

Phase Diagram



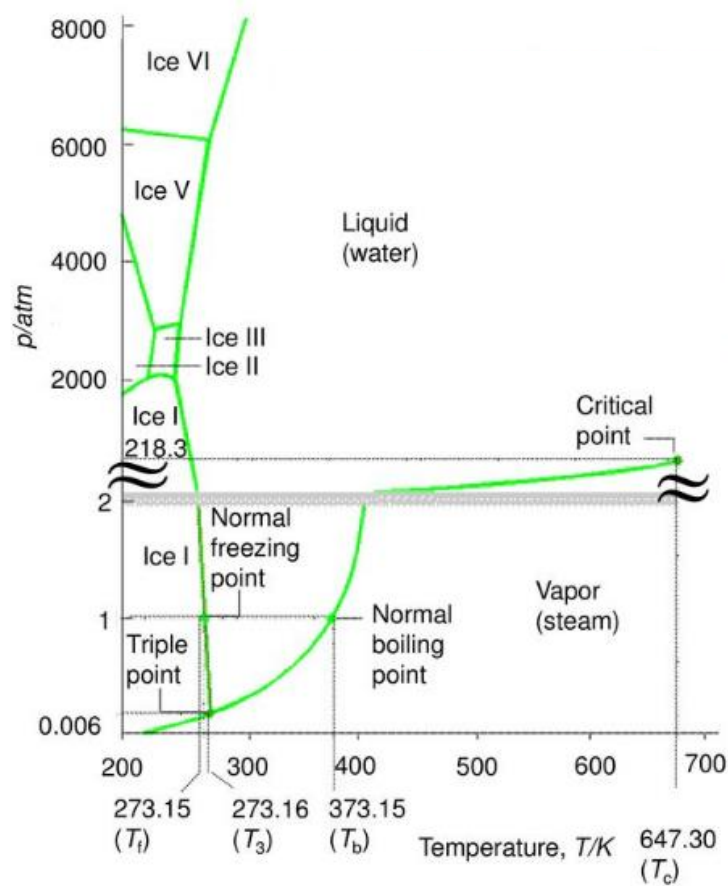
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4B.7)

Revisiting CO₂



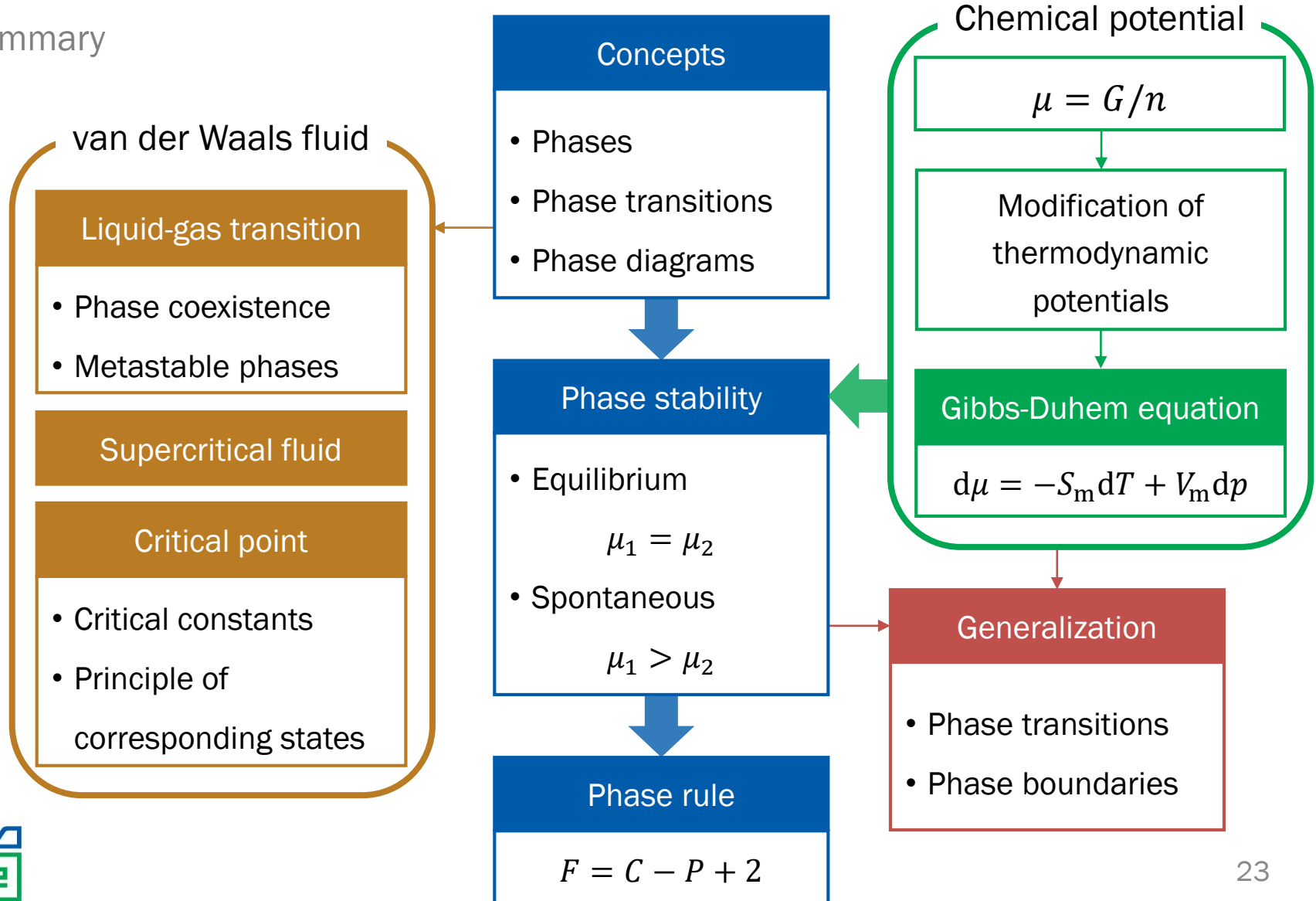
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.8)

Revisiting H₂O



(Image: <https://slideplayer.com/slide/13602761/>)

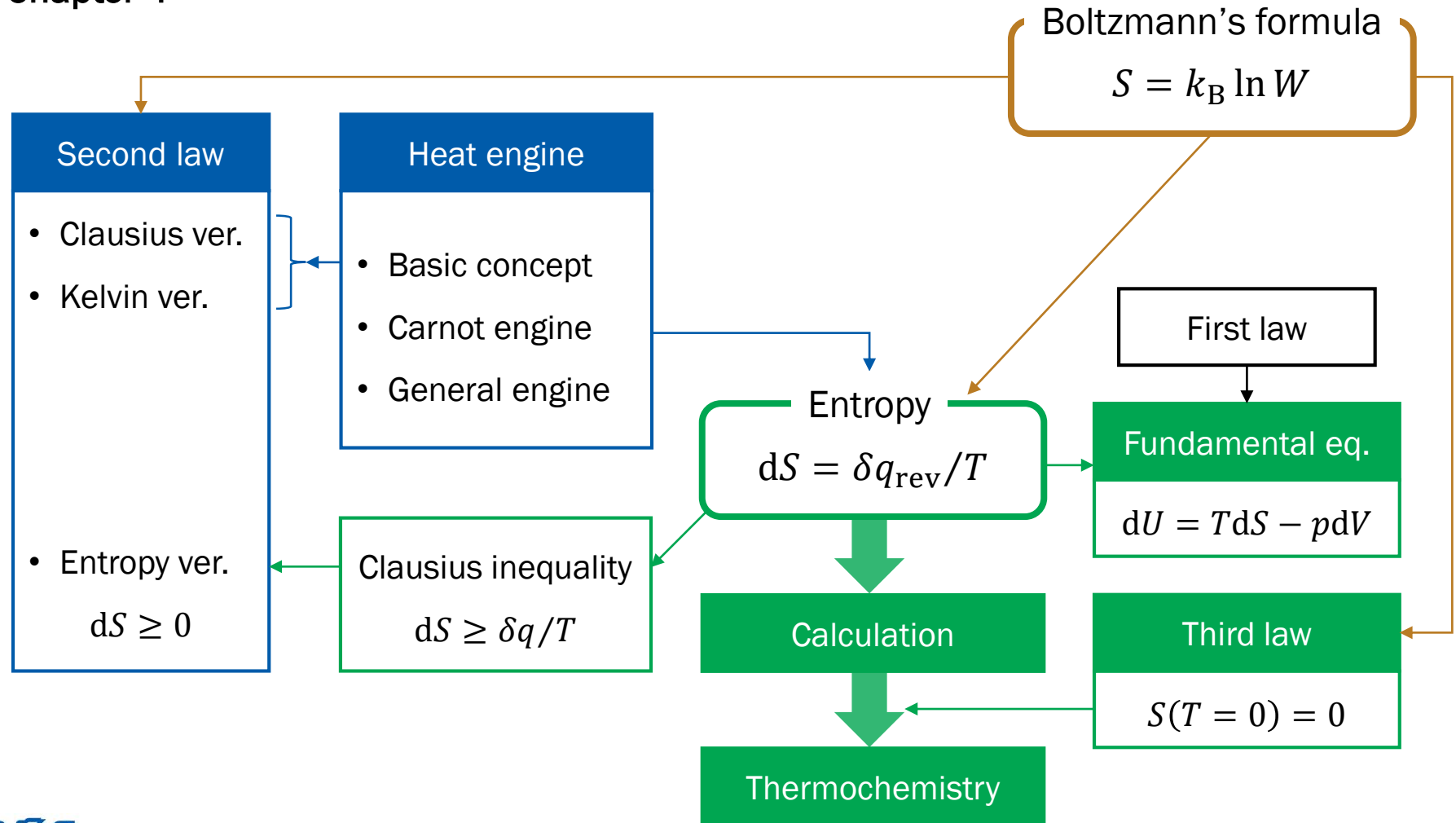
Summary



Review Section



Chapter 4



Concepts and Derivations in Chpt 4

- [D] Entropy calculations
 - Heat transfer
 - Expansion
 - Phase transition
 - Heating
- [C] Nernst heat theorem and the third law
- [C] Thermochemistry of entropy
 - Standard entropy
 - Standard reaction entropy



Concepts and Derivations in Chpt 4

- [C] Entropy as an index of exhaustion
- [C] Boltzmann's formula
- [D] Applications of Boltzmann's formula
- [C] Reinterpretations of the laws
- [C, D] Independent spin model
- [C, D] Shannon entropy

Chapter 5

Thermodynamic potentials

Clausius ineq.

First law

Internal energy

$$U$$

Helmholtz energy

$$A = U - TS$$

Enthalpy

$$H = U + pV$$

Gibbs energy

$$G = H - TS$$

Spontaneity

$$\Delta A_{T,V} \leq 0$$

$$\Delta G_{T,p} \leq 0$$

Maximum work

$$A_T = w_{\max}$$

$$G_{T,p} = w_{\text{add,max}}$$

Differential forms

Applications

- Gibbs-Helmholtz equation
- Maxwell relations
- Thermodynamic equation of state

Thermochemistry

Pressure dependence

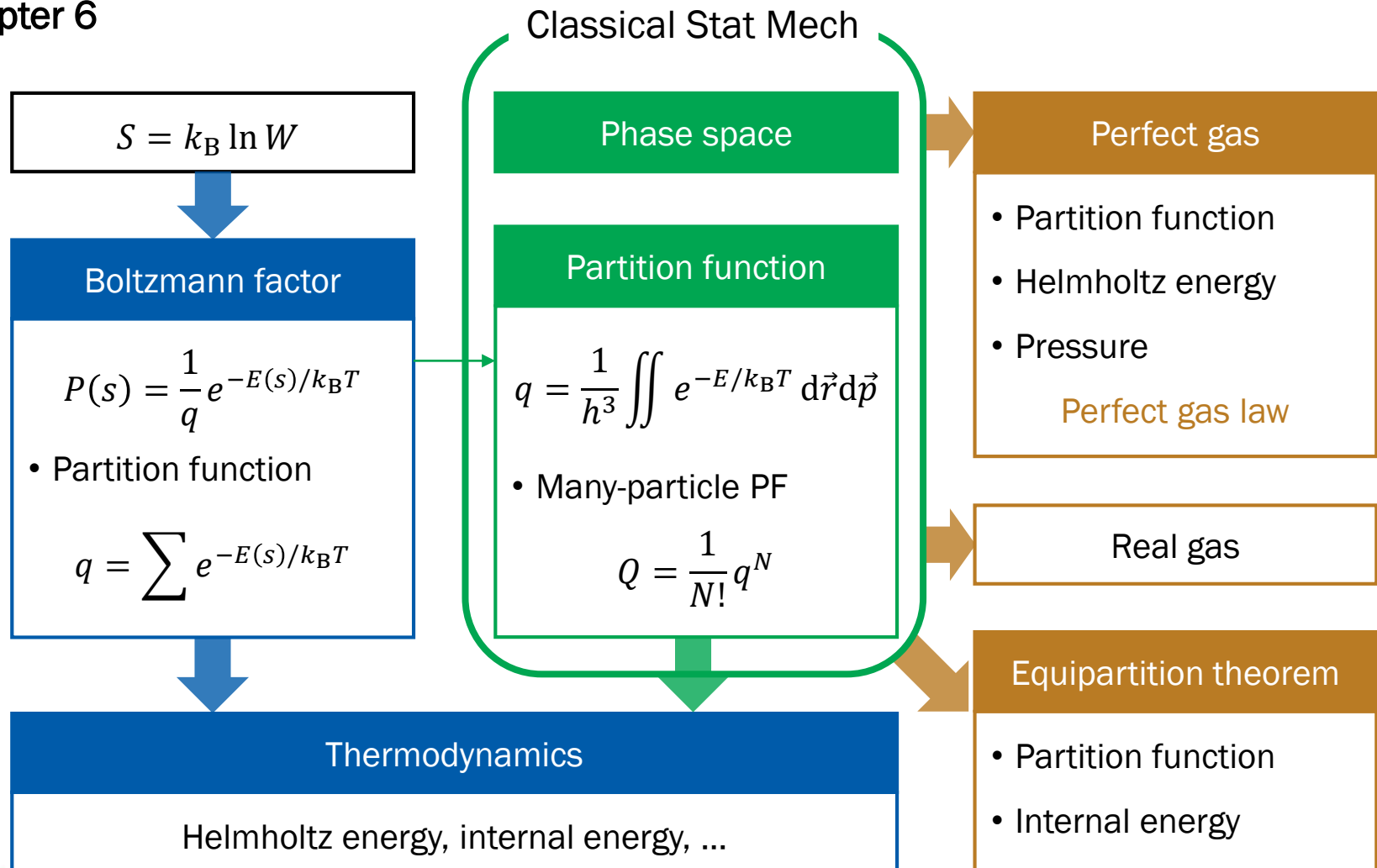


Concepts and Derivations in Chpt 5

- [C] Helmholtz and Gibbs energies
- [D] Spontaneity: Clausius inequality
- [D] Meaning of Helmholtz and Gibbs energies
- [D] Thermodynamic potentials
- [D] Gibbs-Helmholtz equation
- [D] Maxwell relations
- [D] Thermodynamic equation of state
- [D] Pressure dependence of Gibbs energy



Chapter 6



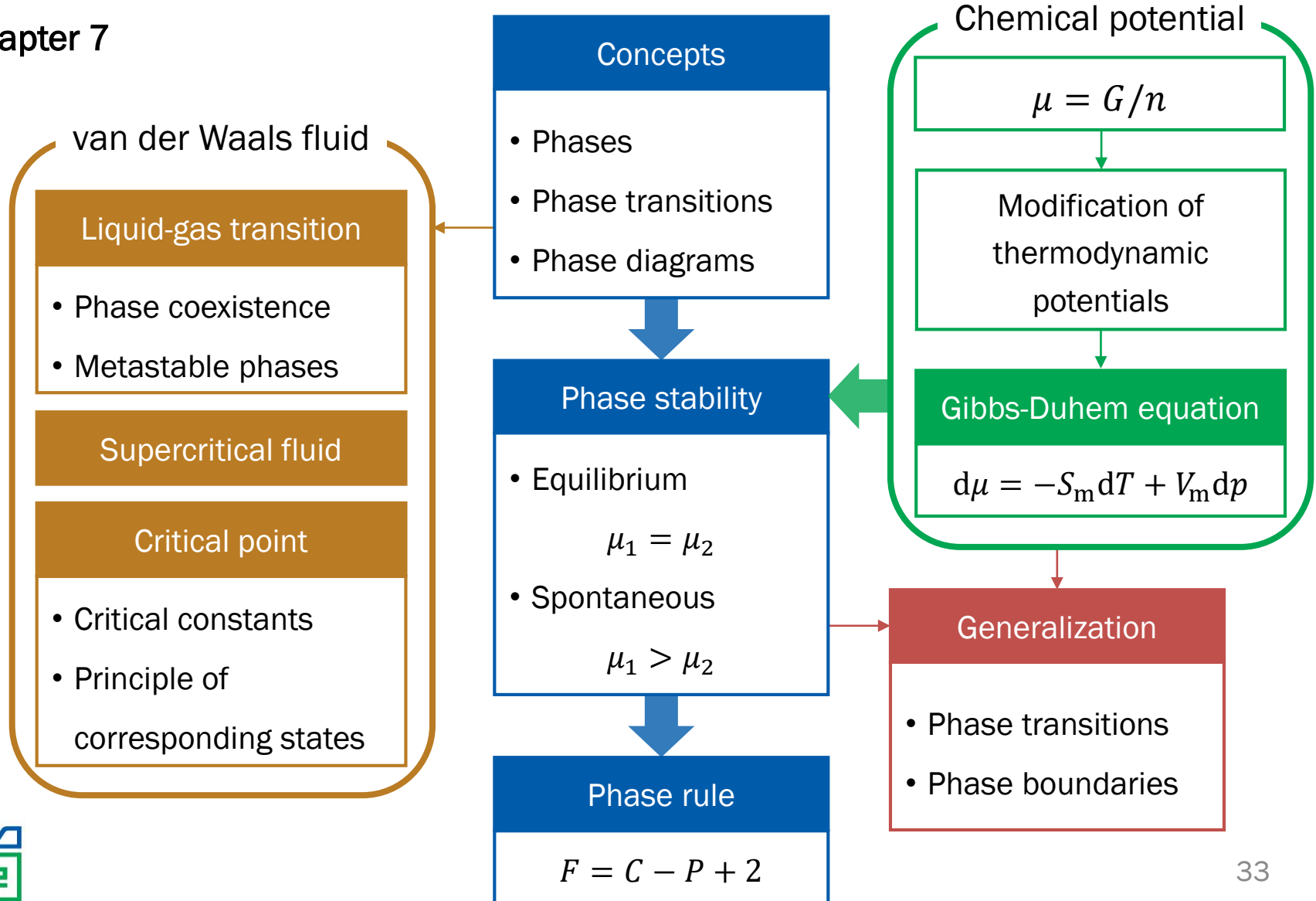
Concepts and Derivations in Chpt 6

- [D] Boltzmann factor
- [C] Partition function
- [D] Partition function and thermodynamics
 - Internal energy
 - Helmholtz energy
 - Other thermodynamic quantities
- [C] Phase space

Concepts and Derivations in Chpt 6

- [C] Partition function of classical system
 - [D] 1-dimensional spring
 - [D] N -particle partition function
 - [C] Indistinguishability
- [D] Perfect gas: perfect gas law
- [C] Real gas: van der Waals equation
- [D] Equipartition theorem

Chapter 7



Concepts and Derivations in Chpt 7

- [C] Phases and phase transitions
- [C, D] Chemical potential
- [D] Rewriting thermodynamic potentials
- [D] Gibbs-Duhem equation
- [D] Phase transition and chemical potential
- [C] Phase diagrams
- [C, D] Phase rule
- [C] Liquid-gas transitions



Concepts and Derivations in Chpt 7

- [C] vdW: Liquid-gas transitions
 - [C] Maxwell construction
 - [D] Lever rule
 - [C] Binodal and spinodal lines
- [C] vdW: Supercritical fluid
- [C] vdW: Critical point
 - [D] Critical constants
 - [D] Critical compression factor
 - [D] Reduced variables
 - [C] Principle of corresponding states



Concepts and Derivations in Chpt 7

- [C] Chemical potentials at transition
 - [D] Use of Gibbs-Duhem equation
- [C] Phase boundaries
 - [D] Clausius equation
 - [D] Clausius-Clapeyron equation
 - [C] Interpretations